

Kartik Kumar

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SUMMARY

- AI Engineer with 1+ year of industry experience delivering LLM-powered systems, multi-agent pipelines, and RAG architectures in enterprise environments. Proficient in Python, LangChain, LangGraph, and cloud platforms including AWS and Azure. Demonstrated ability to take AI solutions from prototype to production, with measurable impact on efficiency, cost reduction, and engineering quality across automotive and technology sectors.

EDUCATION

- Birla Institute of Technology & Science (BITS Pilani)** WILP
M.Tech – Software Engineering *Jan. 2026 – Jan. 2028*
- Thapar Institute of Engineering & Technology** Patiala, Punjab
B.E. – Computer Engineering; CGPA: 8.34/10 *Jul. 2020 – Jun. 2024*

SKILLS SUMMARY

- Machine Learning:** GenAI with LLMs, Fine-tuning with LoRA/QLoRA, NLP, Deep Learning, OpenCV
- AI Frameworks:** LangChain, LangGraph, LangSmith, CrewAI, HuggingFace, MCP, GraphRAG, Agentic Frameworks
- MLOps & Cloud:** AWS, Azure, DataBricks, FastAPI, Docker, MLflow, Pinecone, ChromaDB
- Programming Languages:** Python, C++, SQL

EXPERIENCE

- Magna International Inc.** Bengaluru, India
GenAI Specialist *May 2025 – Present*
 - Architected an **end-to-end Multi-Agent AI system** with orchestrated workflows across specialized agents connected to enterprise knowledge sources via MCP servers, driving an **80% increase in user task efficiency** through intent-driven agent routing and RAG at scale.
 - Delivered a **GenAI-powered quality check system** for engineering requirements and test cases using Azure OpenAI and DataBricks, achieving **85%+ alignment** with human evaluations.
 - Developed an **AI-based consistency check framework** using prompt-tuned LLMs to automate validation of large-scale engineering documentation, reducing manual review overhead.
 - Integrated **MCP (Model Context Protocol)** with OAuth 2.0 and FastAPI, enabling secure, enterprise-grade access to GenAI services and internal business tools.
 - Constructed **agentic chatbots** using CrewAI, LangGraph, and LangChain for enterprise knowledge assistance and workflow automation.
 - Completed a **global onsite assignment in Germany** at Magna's European headquarters, collaborating with cross-functional teams on GenAI solution deployment.**Technical skills:** Python, DataBricks, Azure OpenAI, FastAPI, LangChain, LangGraph, CrewAI, Prompt Engineering, LLM Evaluation.
- Tata Elxsi** Bengaluru, India
GenAI Engineer *Jul. 2024 – May 2025*
 - Streamlined **test case generation and validation** using LLMs, increasing coverage by **25%** and cutting manual QA effort by **40%**.
 - Implemented **Knowledge Graph-based RAG systems**, improving structured data retrieval accuracy and enterprise search relevance.
 - Collaborated onsite with **Mercedes, Magna, Jaguar, and Marelli** to integrate backend GenAI solutions, lifting engineering team productivity across client sites.**Technical skills:** Python, LangChain, GraphRAG, CrewAI, AWS, Azure, Agentic Frameworks.

PUBLICATIONS

An Approach to Integration of Gait and Fingerprint Features for Advanced Biometric Recognition

Mar. 2024

- *IEEE Xplore — 14th Int. Conf. on Cloud Computing, Data Science & Engineering (Confluence)*
<https://ieeexplore.ieee.org/document/10463431>
 - Developed a multimodal biometric recognition framework combining gait and fingerprint features using ResNet-101, VGG-19, and CNN architectures.
 - Achieved **97.12% accuracy** via feature-level fusion and Random Forest classification, strengthening robustness in identity authentication.

PROJECTS

- **Autonomous LinkedIn Content Pipeline:** Launched an AI-powered content automation system that scrapes real-time web trends, generates contextual posts with AI-generated visuals, and publishes on a daily schedule — eliminating manual effort via end-to-end workflow automation.
Tech-Stack: Python, CrewAI, LLMs, Web Scraping, Image Generation, Scheduling. 2025
- **Smart GenAI Model Router:** Formulated a unified API with a multi-signal routing engine that selects LLMs based on task complexity, confidence, risk, and latency — reducing inference costs while improving response speed, with real-time analytics on model usage and cost savings.
Tech-Stack: Python, FastAPI, SQLite, Streamlit, LLMs. 2025
- **Autonomous Cost Control Platform:** Spearheaded an end-to-end cost management system using Python, FastAPI, and Prometheus for real-time financial analytics and anomaly detection. Orchestrated data pipelines, dashboards, and Dockerized deployment to deliver measurable operational savings.
Tech-Stack: Python, FastAPI, SQL, Prometheus, Docker, Data Visualization. 2024
- **Gait Analysis:** Developed a machine learning pipeline to identify individuals from gait patterns, achieving a **20% accuracy improvement** over prior methods.
Tech-Stack: TensorFlow, Keras, Random Forest Classifier, PyQt6. 2023
- **Spotify Billboard Prediction:** Constructed a predictive model to forecast Billboard chart success for Spotify tracks, achieving **85%+ accuracy** and improving recommendation efficiency by 15%.
Tech-Stack: PyCaret, Extra Trees Classifier. 2022
- **AI-Based Surveillance System:** Established a real-time traffic monitoring system using computer vision for traffic density analysis, reducing manual reporting errors by **25%**.
Tech-Stack: OpenCV, YOLO, NumPy, Pandas. 2022
- **Advanced Attendance System:** Implemented a facial recognition attendance system with voice-enabled queries, cutting manual tracking time by **40%**.
Tech-Stack: OpenCV, Face Recognition, Pandas, pyttsx3. 2021

KEY ACHIEVEMENTS & CERTIFICATIONS

- **Certificate of Completion – Generative AI using Large Language Models** (DeepLearning.AI): Completed advanced training on LLM architecture, fine-tuning, and deployment best practices.
- **Course Certificate – Prompt Engineering for ChatGPT** (Vanderbilt University): Demonstrated proficiency in systematic prompt design for production AI applications.
- **College Winner – Tata Teleport 2023** (Dec. 2023): Secured first place in a national-level competition conducted by Tata Elxsi.
- **Top 5 – Kaggle Data Scientist Competition** (Numer.ai Stocks Dataset): Ranked in the top 5 in a competitive stock market prediction challenge.
- **T.I.E.T Merit Scholarship Recipient** (2021): Awarded for academic excellence during the 2020–2021 academic year.